

Week 5

Healthcare Data Cleaning & Data Modeling

Power Query Data Cleaning and Star-Schema Modeling in Power BI

Dataset: Healthcare Dataset

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1. Task Overview

1.1 Objectives

- Import and inspect patient records, admission types, and billing fields in Power Query.
- Handle null values in diagnostic codes and standardize categorical text attributes.
- Create derived conditional columns (e.g., Age Group buckets: 0–18, 19–35, 36–60, 60+).
- Structure dimension tables (Dim_Patient, Dim_Admission) and fact tables (Fact_Billing).

1.2 Tasks & Deliverables

- Group patient ages into logical demographic categories via Power Query.
- Standardize admission types (Emergency, Elective, Urgent).
- Set up star-schema model relationships between patient dimensions and admission facts.
- Output a cleaned data model ready for clinical analytics.

2. Importing and Inspecting the Dataset

The healthcare_dataset was loaded into Power Query and column data types were reviewed and set (Name, Age, Gender, Blood Type, Medical Condition, Date of Admission, Doctor, and related fields), giving a first look at data quality across all 15 columns.

	Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor
1	Bobby Jackson	30	Male	B-	Cancer	31-01-2024	Matthew Smi
2	Leslie Terry	62	Male	A+	Obesity	20-08-2019	Samantha Da
3	Dakny Smith	76	Female	A-	Obesity	22-09-2022	Tiffany Mitch
4	andrew watts	28	Female	O+	Diabetes	18-11-2020	Kevin Wells
5	adrienne bell	43	Female	AB+	Cancer	19-09-2022	Kathleen Han
6	EMILY JOHNSON	36	Male	A+	Asthma	20-12-2023	Taylor Newto
7	edward EDWARDS	21	Female	AB-	Diabetes	03-12-2020	Kelly Olson
8	CHRISTINA MARTINEZ	20	Female	A+	Cancer	28-12-2021	Susanne Thor
9	JASMIN Aguilar	82	Male	AB+	Asthma	01-07-2020	Daniel Fergus
10	CHRISTOPHER BERG	58	Female	AB-	Cancer	23-05-2021	Heather Day
11	michelle danielis	72	Male	O+	Cancer	19-04-2020	John Duncan
12	aaron MARTINEZ	38	Female	A-	Hypertension	13-08-2023	Douglas May
13	connor HANSEN	75	Female	A+	Diabetes	12-12-2019	Kenneth Fletc
14	robert blauer	68	Female	AB+	Asthma	22-05-2020	Theresa Freee
15	BROCKE brady	44	Female	AB+	Cancer	08-10-2021	Roberta Stew
16	MS. nAtalie gAMble	46	Female	AB-	Obesity	01-01-2023	Maria Dough
17	haley perkins	63	Female	A+	Arthritis	23-06-2020	Erica Spencer
18	mrs. jamie CAMPBELL	38	Male	AB-	Obesity	08-03-2020	Justin Kim
19	LUKE BURGESS	34	Female	A-	Hypertension	04-03-2021	Justin Moore
20	DANIEL schmidt	63	Male	B+	Asthma	15-11-2022	Denise Gallow
21	TIMOTHY burns	67	Female	A-	Asthma	28-06-2023	Krista Smith
22	CHRISTOPHER BRIGHT	48	Male	B+	Asthma	21-01-2020	Gregory Smit
23	KATHRYN Stewart	58	Female	O+	Arthritis	12-05-2022	Vanessa New
24	DR. ELIEN thompson	59	Male	A+	Asthma	02-08-2021	Donna Marti
25							

3. Standardizing Text Attributes

As part of standardizing categorical and text attributes, the Name column was transformed to uppercase text using the Format > Uppercase transformation.

Table.TransformColumns(*"Changed column type",{"Name", Text.Upper, type text})

Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results
1	18856.28131	328	Urgent	02-02-2024	Paracetamol	Normal
2	33643.32729	265	Emergency	26-08-2019	Ibuprofen	Inconclusive
3	27955.09608	205	Emergency	07-10-2022	Aspirin	Normal
4	37908.78241	450	Elective	18-12-2020	Ibuprofen	Abnormal
5	14238.31781	458	Urgent	09-10-2022	Penicillin	Abnormal
6 re	48145.11095	389	Urgent	24-12-2023	Ibuprofen	Normal
7	19580.87234	389	Emergency	15-11-2020	Paracetamol	Inconclusive
8	45820.46272	277	Emergency	07-01-2022	Paracetamol	Inconclusive
9	50119.22279	316	Elective	14-07-2020	Aspirin	Abnormal
10 re	19784.63106	249	Elective	22-06-2021	Paracetamol	Inconclusive
11	12576.79561	394	Urgent	22-04-2020	Paracetamol	Normal
12	7999.58688	288	Urgent	05-09-2023	Lipitor	Inconclusive
13	43282.28336	134	Emergency	28-12-2019	Penicillin	Abnormal
14 re	33207.70663	309	Urgent	19-06-2020	Lipitor	Normal
15 re	40701.59923	182	Urgent	13-10-2021	Paracetamol	Normal
16	12263.35743	465	Elective	11-01-2023	Aspirin	Inconclusive
17 re	24499.8479	114	Elective	14-07-2020	Paracetamol	Normal
18	17440.46544	449	Urgent	02-04-2020	Paracetamol	Abnormal
19	18843.02302	260	Elective	14-03-2021	Aspirin	Abnormal
20	23762.20358	465	Elective	22-11-2022	Penicillin	Normal
21	42.51458855	115	Elective	02-07-2023	Aspirin	Normal
22	17695.91162	295	Urgent	09-02-2020	Lipitor	Normal
23	5998.102908	327	Urgent	10-06-2022	Lipitor	Inconclusive
24	25250.05243	119	Urgent	12-08-2021	Lipitor	Inconclusive
25	53311.50243	185	Emergency	05-02-2020	Paracetamol	Inconclusive

15 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 08:46

4. Creating the Age Group Conditional Column

A conditional column, Age Group, was created using Power Query's Add Conditional Column tool. Patients were bucketed as Child (≤ 18), Young (19–35), Adult (36–60), and Senior (60+), directly fulfilling the age-grouping deliverable.

Table.TransformColumns(*"Changed column type",{"Name", Text.Upper, type text})

Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results
1	18856.28131	328	Urgent	02-02-2024	Paracetamol	Normal
2	33643.32729	265	Emergency	26-08-2019	Ibuprofen	Inconclusive
3	27955.09608	205	Emergency	07-10-2022	Aspirin	Normal
4	37908.78241	450	Elective	18-12-2020	Ibuprofen	Abnormal
5	14238.31781	458	Urgent	09-10-2022	Penicillin	Abnormal
6 re	48145.11095	389	Urgent	24-12-2023	Ibuprofen	Normal
7	19580.87234	389	Emergency	15-11-2020	Paracetamol	Inconclusive
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9	50119.22279	316	Elective	14-07-2020	Aspirin	Abnormal
10 re	19784.63106	249	Elective	22-06-2021	Paracetamol	Inconclusive
11	12576.79561	394	Urgent	22-04-2020	Paracetamol	Normal
12	7999.58688	288	Urgent	05-09-2023	Lipitor	Inconclusive
13	43282.28336	134	Emergency	28-12-2019	Penicillin	Abnormal
14 re	33207.70663	309	Urgent	19-06-2020	Lipitor	Normal
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20	23762.20358	465	Elective	22-11-2022	Penicillin	Normal
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22	17695.91162	295	Urgent	09-02-2020	Lipitor	Normal
23	5998.102908	327	Urgent	10-06-2022	Lipitor	Inconclusive
24	25250.05243	119	Urgent	12-08-2021	Lipitor	Inconclusive
25	53311.50243	185	Emergency	05-02-2020	Paracetamol	Inconclusive

15 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 08:46

5. Formatting Categorical Columns

Admission Type text was set to Proper Case and the new Age Group column was formatted with Capitalize Each Word, before applying and closing the query editor to load the cleaned table.

The screenshot shows the Power Query Editor interface. The main table, 'healthcare_dataset', has columns: Room Number, Admission Type, Discharge Date, Medication, Test Results, and Age Group. The 'Age Group' column is highlighted. In the 'Applied Steps' pane on the right, the step 'Capitalized Each Word' is selected. The formula bar shows the transformation: `Table.TransformColumns(#Added Conditional Column,{"Admission Type", Text.Proper, type text})`. The status bar at the bottom indicates '16 COLUMNS, 999+ ROWS'.

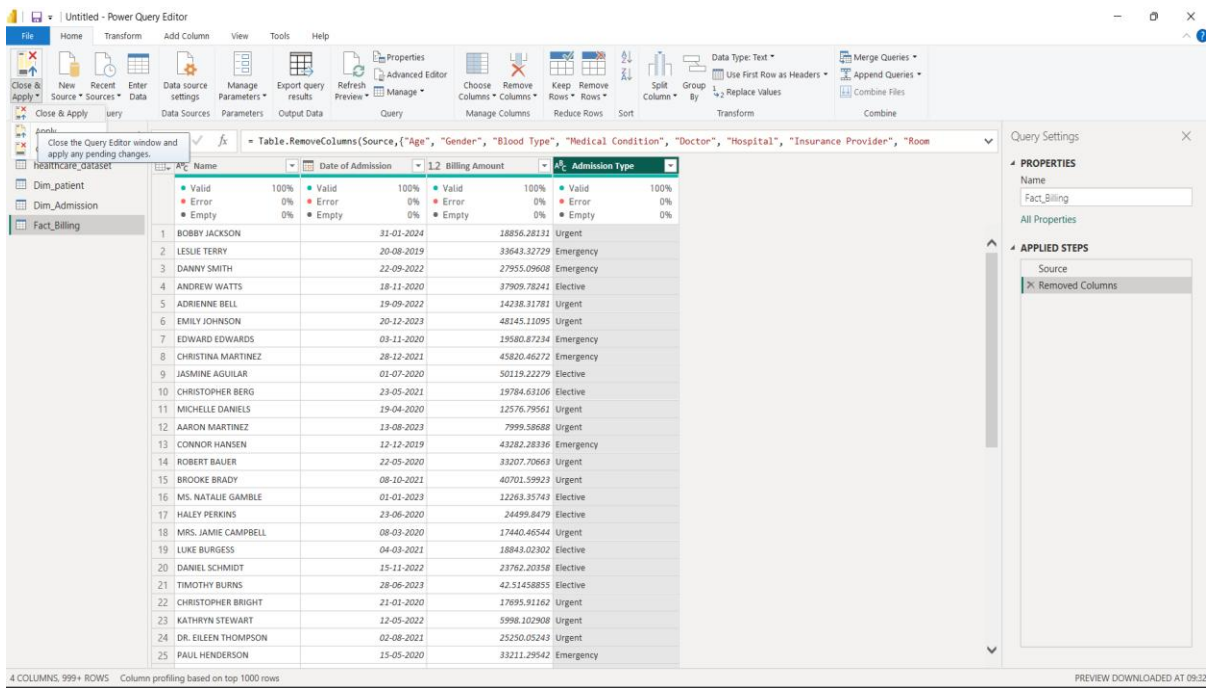
6. Building the Dim_Admission Dimension Table

A dedicated Dim_Admission query was created by removing patient-demographic columns, retaining only admission-related fields such as Date of Admission, Room Number, Admission Type, and Discharge Date.

The screenshot shows the Power Query Editor interface. The main table, 'Dim_Admission', has columns: Name, Medical Condition, Date of Admission, Room Number, Admission Type, and Discharge Date. The 'Removed Columns' step is selected in the 'Applied Steps' pane. The formula bar shows the transformation: `Table.RemoveColumns(Source,{"Age", "Gender", "Blood Type", "Billing Amount", "Age Group", "Doctor", "Hospital", "Insurance Provider", "Medication", "Test Results", "Age Group"})`. The status bar at the bottom indicates '6 COLUMNS, 999+ ROWS'.

7. Building the Billing Table

The Fact_Billing query was created by removing non-billing columns, leaving Name, Date of Admission, Billing Amount, and Admission Type as the core billing facts.



	Name	Date of Admission	Billing Amount	Admission Type
1	BOBBY JACKSON	31-01-2024	18856.28131	Urgent
2	LESLIE TERRY	20-08-2019	33643.32729	Emergency
3	DANNY SMITH	22-09-2022	27955.09608	Emergency
4	ANDREW WATTS	18-11-2020	37909.78241	Elective
5	ADRIENNE BELL	19-09-2022	14218.31781	Urgent
6	EMILY JOHNSON	20-12-2023	48145.11095	Urgent
7	EDWARD EDWARDS	03-11-2020	19580.87234	Emergency
8	CHRISTINA MARTINEZ	28-12-2021	45820.46272	Emergency
9	JASMINE AGUILAR	01-07-2020	50119.22279	Elective
10	CHRISTOPHER BERG	23-05-2021	19784.63106	Elective
11	MICHELLE DANIELS	19-04-2020	12576.79561	Urgent
12	AARON MARTINEZ	13-08-2023	7999.58688	Urgent
13	CONNOR HANSEN	13-12-2019	43282.28336	Emergency
14	ROBERT BAUER	22-05-2020	33207.70663	Urgent
15	BROOKE BRADY	08-10-2021	40701.59923	Urgent
16	MS. NATALIE GAMBLE	01-01-2023	12263.35743	Elective
17	HALEY PERKINS	23-06-2020	24499.8479	Elective
18	MRS. JAMIE CAMPBELL	08-03-2020	17440.46544	Urgent
19	LUKE BURGESS	04-03-2021	18843.02302	Elective
20	DANIEL SCHMIDT	15-11-2022	23762.20358	Elective
21	TIMOTHY BURNS	28-06-2023	42.51458855	Elective
22	CHRISTOPHER BRIGHT	21-01-2020	17695.91162	Urgent
23	KATHRYN STEWART	13-05-2022	5998.102908	Urgent
24	DR. EILEEN THOMPSON	02-08-2021	25250.05243	Urgent
25	PAUL HENDERSON	15-05-2020	33211.29542	Emergency

8. Loading the Initial Model into Power BI

The four queries — healthcare_dataset, Dim_patient, Fact_Billing, and Dim_Admission — were loaded into Power BI Desktop, ready for relationship modeling.

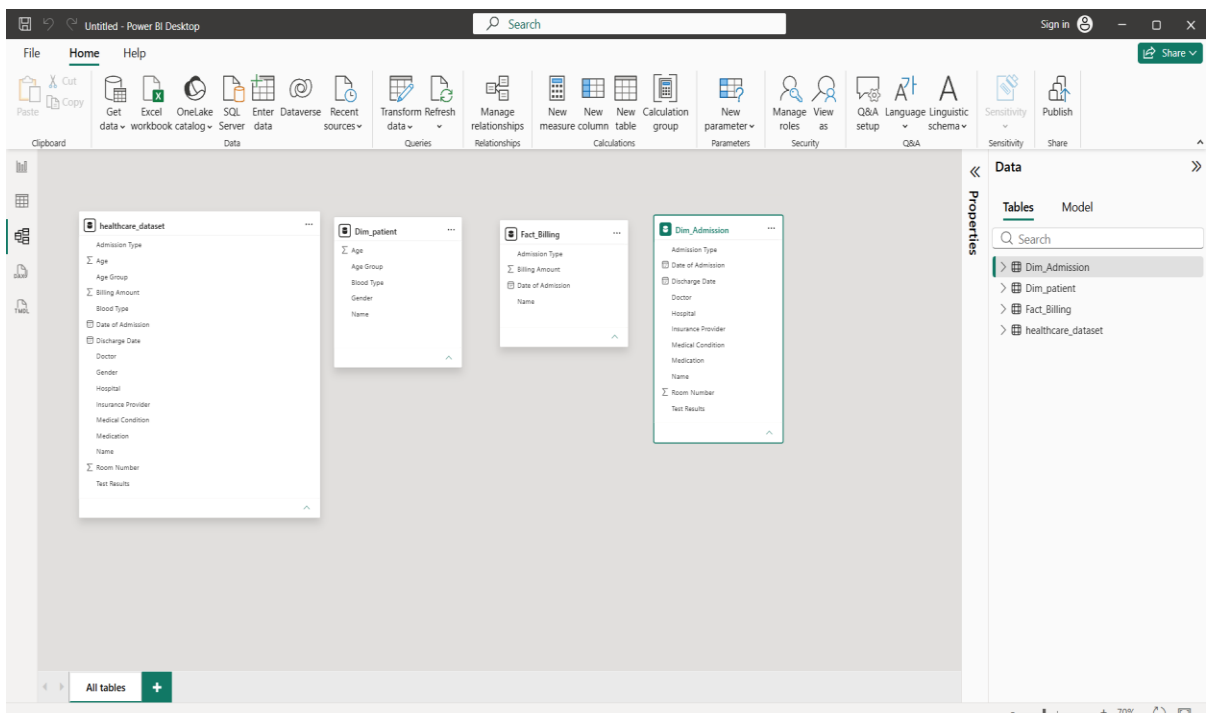
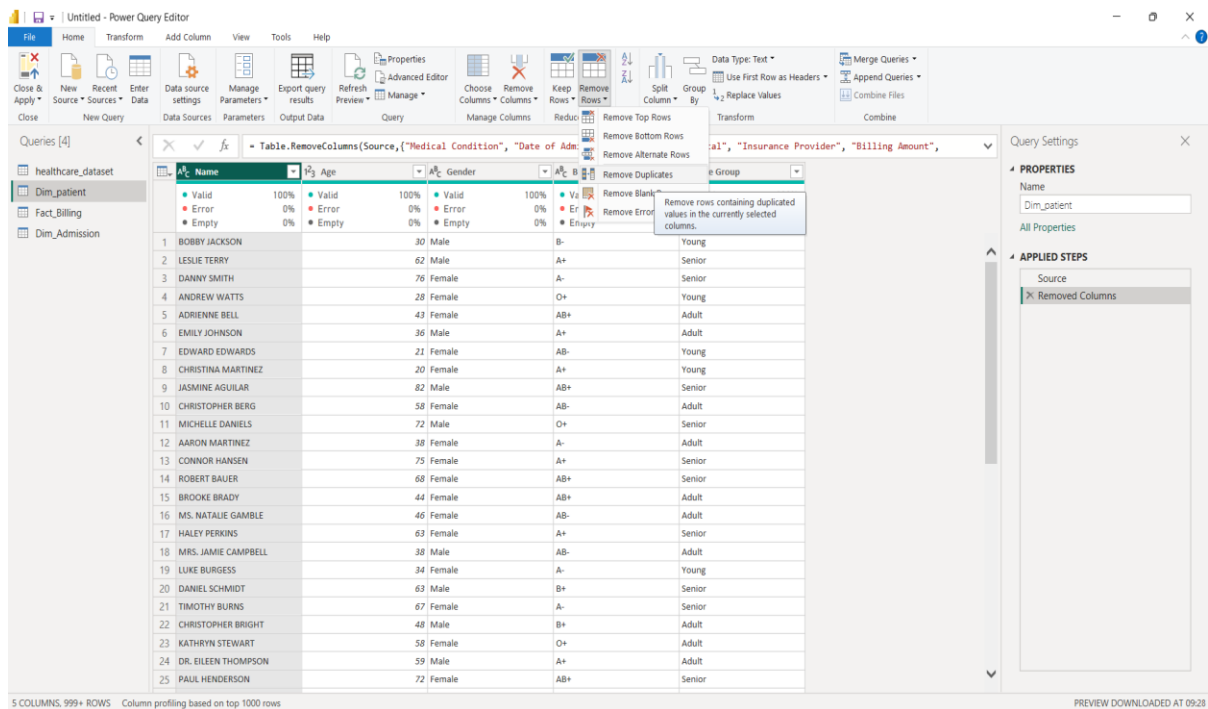


Table	Columns
healthcare_dataset	Admission Type, Age Group, Billing Amount, Date of Admission, Discharge Date, Doctor, Gender, Hospital, Insurance Provider, Medical Condition, Medication, Name, Room Number, Test Results
Dim_patient	Age Group, Blood Type, Gender, Name
Fact_Billing	Admission Type, Billing Amount, Date of Admission, Name
Dim_Admission	Admission Type, Date of Admission, Discharge Date, Doctor, Hospital, Insurance Provider, Medical Condition, Medication, Name, Room Number, Test Results

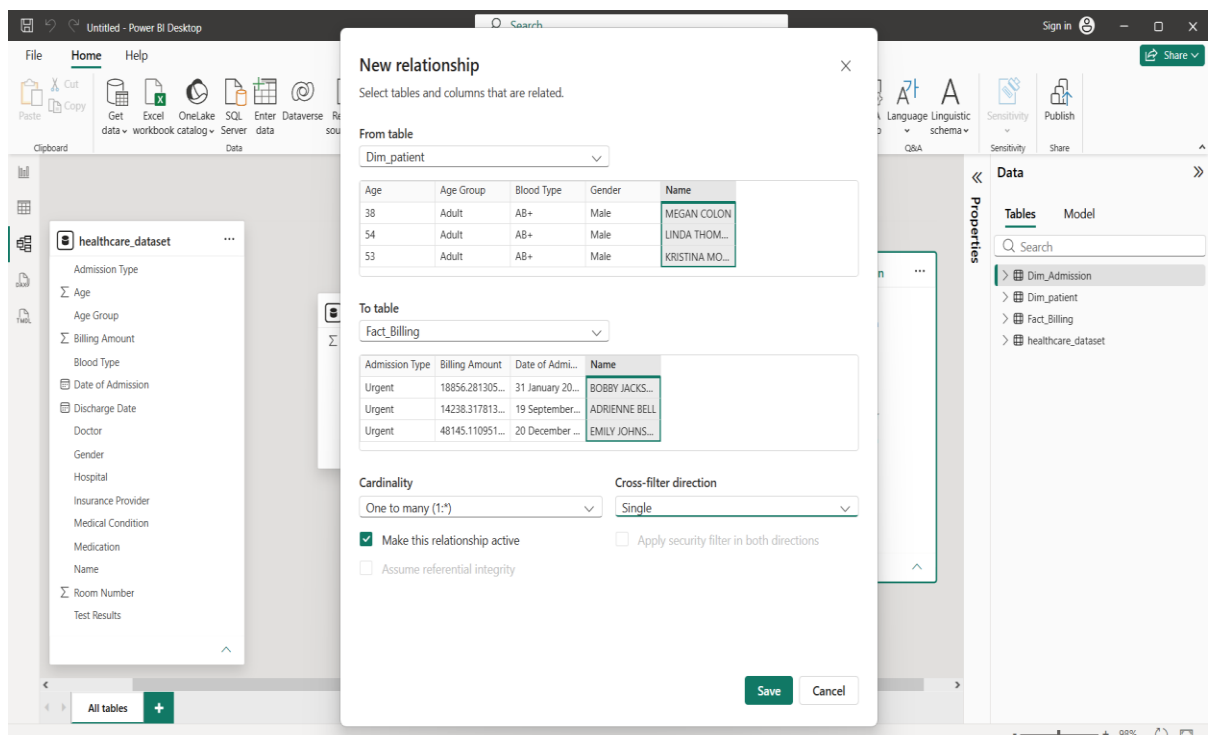
9. Removing Duplicate Patients

Duplicate rows were removed from the Dim_patient query using Remove Rows > Remove Duplicates, ensuring each patient appears only once in the dimension table.



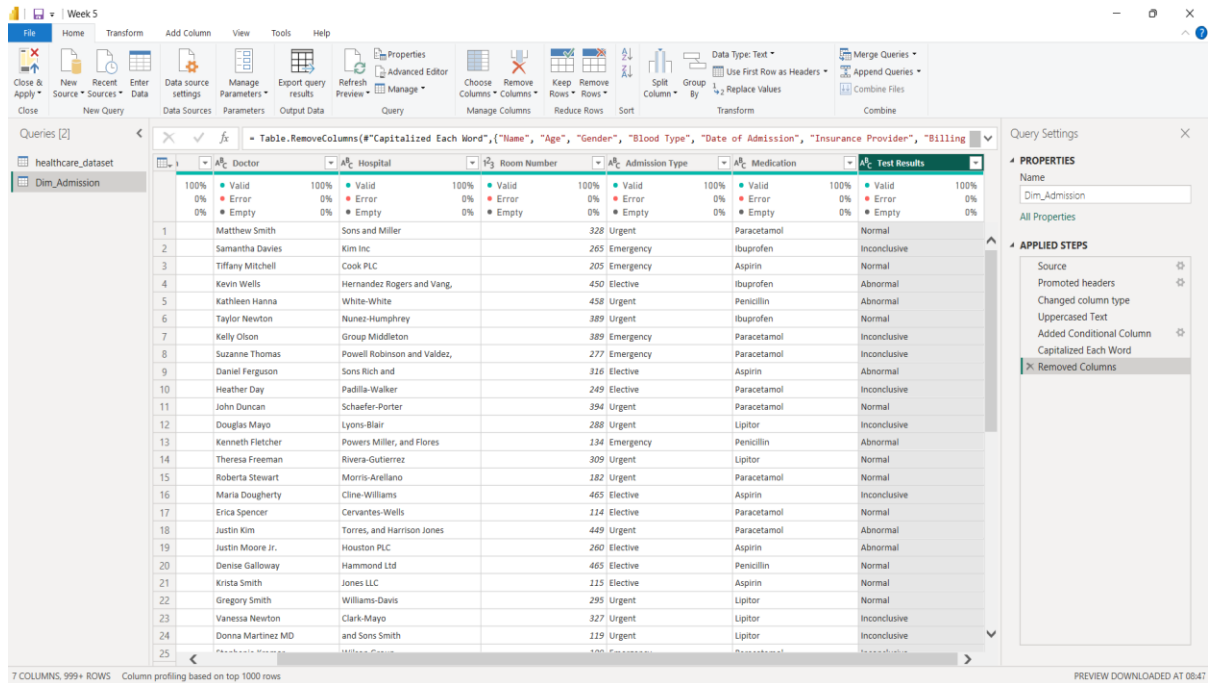
10. Creating the Patient–Billing Relationship

A one-to-many relationship was defined between Dim_patient and Fact_Billing on the Name field, with single-direction cross-filtering, establishing the first link of the star schema.



11. Refining the Dim_Admission Table

In a following session, additional columns (Name, Age, Gender, Blood Type, Date of Admission, Insurance Provider, Billing Amount) were removed from Dim_Admission, leaving a clean set of admission attributes: Doctor, Hospital, Room Number, Admission Type, Medication, and Test Results.

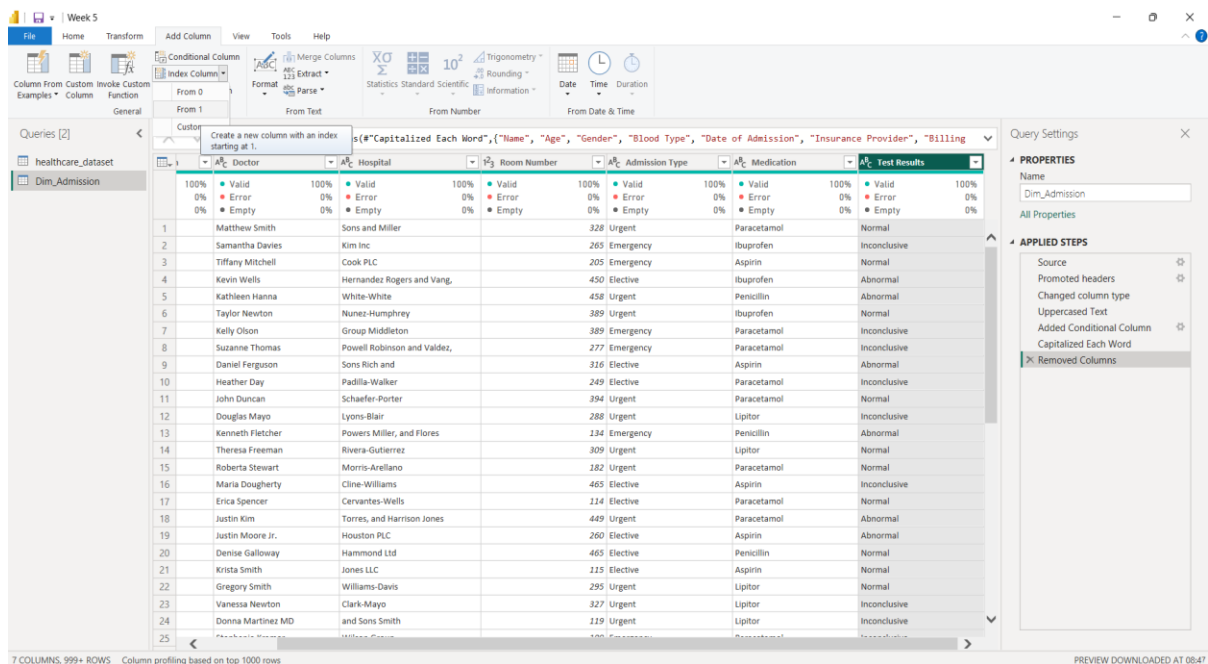


The screenshot shows the Power BI Desktop interface with the 'Dim_Admission' table loaded. The table has 7 columns: Doctor, Hospital, Room Number, Admission Type, Medication, and Test Results. The data is displayed in a grid with 25 rows. The right-hand pane shows the 'Query Settings' for 'Dim_Admission', with the 'Properties' tab selected, showing the table name and the 'Applied Steps' list.

	Doctor	Hospital	Room Number	Admission Type	Medication	Test Results
1	Matthew Smith	Sons and Miller	328	Urgent	Paracetamol	Normal
2	Samantha Davies	Kim Inc	265	Emergency	ibuprofen	Inconclusive
3	Tiffany Mitchell	Cook PLC	205	Emergency	Aspirin	Normal
4	Kevin Wells	Hernandez Rogers and Vang,	450	Elective	ibuprofen	Abnormal
5	Kathleen Hanna	White-White	458	Urgent	Penicillin	Abnormal
6	Taylor Newton	Nunez-Humphrey	389	Urgent	ibuprofen	Normal
7	Kelly Olson	Group Middleton	389	Emergency	Paracetamol	Inconclusive
8	Suzanne Thomas	Powell Robinson and Valdez,	277	Emergency	Paracetamol	Inconclusive
9	Daniel Ferguson	Sons Rich and	316	Elective	Aspirin	Abnormal
10	Heather Day	Padilla-Walker	249	Elective	Paracetamol	Inconclusive
11	John Duncan	Schaefer-Porter	394	Urgent	Paracetamol	Normal
12	Douglas Mayo	Lyons-Blair	288	Urgent	Lipitor	Inconclusive
13	Kenneth Fletcher	Powers Miller, and Flores	134	Emergency	Penicillin	Abnormal
14	Theresa Freeman	Rivera-Gutierrez	309	Urgent	Lipitor	Normal
15	Roberta Stewart	Morris-Arellano	182	Urgent	Paracetamol	Normal
16	Maria Dougherty	Cline-Williams	465	Elective	Aspirin	Inconclusive
17	Erica Spencer	Cervantes-Wells	114	Elective	Paracetamol	Normal
18	Justin Kim	Torres, and Harrison Jones	449	Urgent	Paracetamol	Abnormal
19	Justin Moore Jr.	Houston PLC	260	Elective	Aspirin	Abnormal
20	Denise Galloway	Hammond Ltd	465	Elective	Penicillin	Normal
21	Krista Smith	Jones LLC	115	Elective	Aspirin	Normal
22	Gregory Smith	Williams-Davis	295	Urgent	Lipitor	Normal
23	Vanessa Newton	Clark-Mayo	327	Urgent	Lipitor	Inconclusive
24	Donna Martinez MD	and Sons Smith	119	Urgent	Lipitor	Inconclusive
25	Franklin Brown	Wilson Group	400	Emergency	Paracetamol	Inconclusive

12. Adding a Surrogate Admission ID

An Index Column starting from 1 was added via Add Column > Index Column, creating a surrogate Admission ID to serve as the primary key of the Dim_Admission table.



The screenshot shows the Power BI Desktop interface with the 'Dim_Admission' table. A new 'Index' column has been added at the beginning of the table, starting from 1. The table now has 8 columns: Index, Doctor, Hospital, Room Number, Admission Type, Medication, and Test Results. The data is displayed in a grid with 25 rows. The right-hand pane shows the 'Query Settings' for 'Dim_Admission', with the 'Properties' tab selected, showing the table name and the 'Applied Steps' list.

Index	Doctor	Hospital	Room Number	Admission Type	Medication	Test Results
1	Matthew Smith	Sons and Miller	328	Urgent	Paracetamol	Normal
2	Samantha Davies	Kim Inc	265	Emergency	ibuprofen	Inconclusive
3	Tiffany Mitchell	Cook PLC	205	Emergency	Aspirin	Normal
4	Kevin Wells	Hernandez Rogers and Vang,	450	Elective	ibuprofen	Abnormal
5	Kathleen Hanna	White-White	458	Urgent	Penicillin	Abnormal
6	Taylor Newton	Nunez-Humphrey	389	Urgent	ibuprofen	Normal
7	Kelly Olson	Group Middleton	389	Emergency	Paracetamol	Inconclusive
8	Suzanne Thomas	Powell Robinson and Valdez,	277	Emergency	Paracetamol	Inconclusive
9	Daniel Ferguson	Sons Rich and	316	Elective	Aspirin	Abnormal
10	Heather Day	Padilla-Walker	249	Elective	Paracetamol	Inconclusive
11	John Duncan	Schaefer-Porter	394	Urgent	Paracetamol	Normal
12	Douglas Mayo	Lyons-Blair	288	Urgent	Lipitor	Inconclusive
13	Kenneth Fletcher	Powers Miller, and Flores	134	Emergency	Penicillin	Abnormal
14	Theresa Freeman	Rivera-Gutierrez	309	Urgent	Lipitor	Normal
15	Roberta Stewart	Morris-Arellano	182	Urgent	Paracetamol	Normal
16	Maria Dougherty	Cline-Williams	465	Elective	Aspirin	Inconclusive
17	Erica Spencer	Cervantes-Wells	114	Elective	Paracetamol	Normal
18	Justin Kim	Torres, and Harrison Jones	449	Urgent	Paracetamol	Abnormal
19	Justin Moore Jr.	Houston PLC	260	Elective	Aspirin	Abnormal
20	Denise Galloway	Hammond Ltd	465	Elective	Penicillin	Normal
21	Krista Smith	Jones LLC	115	Elective	Aspirin	Normal
22	Gregory Smith	Williams-Davis	295	Urgent	Lipitor	Normal
23	Vanessa Newton	Clark-Mayo	327	Urgent	Lipitor	Inconclusive
24	Donna Martinez MD	and Sons Smith	119	Urgent	Lipitor	Inconclusive
25	Franklin Brown	Wilson Group	400	Emergency	Paracetamol	Inconclusive

13. Merging Billing with Patient Data

The Billing query was merged with Dim_patient using a Left Outer join, bringing patient attributes such as Age, Gender, Blood Type, and Age Group into the billing fact table.

The screenshot shows the Power BI Desktop interface with the 'Merge' dialog box open. The dialog is configured to merge the 'Billing' table with the 'Dim_patient' table. The 'Billing' table has columns: Billing Amount, Room Number, Admission Type, Discharge Date, Medication, Test Results, and Age Group. The 'Dim_patient' table has columns: Patient ID, Name, Age, Gender, Blood Type, and Age Group. The 'Join Kind' is set to 'Left Outer (all from first, matching from second)'. The dialog indicates that the selection matches 40235 rows from the first table. The 'Applied Steps' pane on the right shows 'Removed Duplicates'.

14. Expanding the Merged Patient Columns

The merged Dim_patient column was expanded within the Billing query, appending Name, Age, and Gender fields alongside the existing billing data.

The screenshot shows the Power BI Desktop interface with the 'Billing' query expanded. The query is now a table with columns: Admission Type, Discharge Date, Medication, Test Results, Age Group, and Dim_patient. The Dim_patient column has been expanded to show Name, Age, and Gender. The 'Applied Steps' pane on the right shows 'Merged Queries'.

15. Merging Billing with Admission Data

A second merge joined Billing with Dim_Admission, linking each billing record to its corresponding admission details and completing the second fact-to-dimension relationship.

The screenshot shows the Power BI Desktop interface with the 'Merge' dialog box open. The dialog is titled 'Merge' and contains the following sections:

- Select a table and matching columns to create a merged table.**
- Billing** table: Columns include Patient ID, Billing Amount, Room Number, Admission Type, Discharge Date, Medication, Test Results, and Age Group.
- Dim_Admission** table: Columns include Doctor, Hospital, Room Number, Admission Type, Medication, Test Results, and Age Group.
- Join Kind:** Set to 'Left Outer (all from first, matching from second)'.
- Fuzzy matching options:** Includes 'Use fuzzy matching to perform the merge' and 'Estimating matches based on data previews'.

The 'Properties' pane on the right shows the 'Query Settings' for the 'Billing' query, including 'Name', 'Source', 'Promoted headers', 'Changed column type', 'Uppercased Text', 'Added Conditional Column', 'Capitalized Each Word', 'Duplicated Column', 'Removed Columns', 'Removed Duplicates', 'Merged Queries', 'Expanded Dim_patient', and 'Reordered Columns'.

16. Selecting the Admission Foreign Key

From the merged Dim_Admission table, only the Admission ID column was expanded into Billing, establishing a clean foreign key between Fact_Billing and Dim_Admission without duplicating admission attributes.

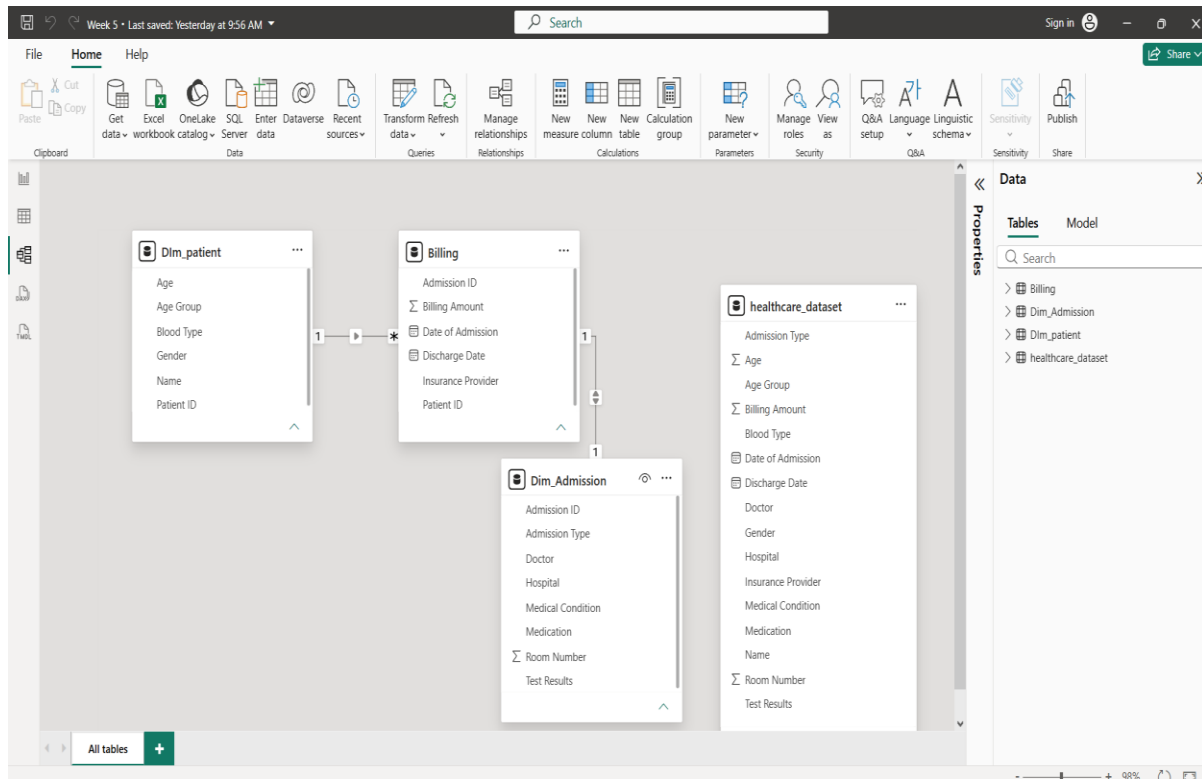
The screenshot shows the Power BI Desktop interface with the 'Expand' dialog box open. The dialog is titled 'Expand' and contains the following sections:

- Search Columns to Expand:** A search bar with the text 'Admission ID'.
- Expand** button: A button to expand the selected column.
- Expand** button: A button to expand the selected column.
- Expand** button: A button to expand the selected column.

The 'Properties' pane on the right shows the 'Query Settings' for the 'Billing' query, including 'Name', 'Source', 'Promoted headers', 'Changed column type', 'Uppercased Text', 'Added Conditional Column', 'Capitalized Each Word', 'Duplicated Column', 'Removed Columns', 'Removed Duplicates', 'Merged Queries', 'Expanded Dim_patient', and 'Reordered Columns'.

17. Final Data Model

The completed star-schema model shows Billing acting as the central fact table, connected to the Dim_patient and Dim_Admission dimension tables, alongside the original healthcare_dataset query — ready to support the interactive Power BI report.



18. Conclusion

The raw healthcare dataset was cleaned and restructured in Power Query: text fields were standardized, an Age Group conditional column was derived, duplicate patient records were removed, and the data was split into a Dim_patient dimension, a Dim_Admission dimension (with a surrogate Admission ID), and a Fact_Billing fact table. These tables were linked in Power BI Desktop through one-to-many relationships, producing a clean star-schema model that is ready for building an interactive clinical analytics report.